



## Southeast University

Department of Computer Science and Engineering

### LAB Report

Course Name: Computer Networking LAB

Course Code: CSE342.2

Experiment No.: 01

Experiment Name: Basic Router Interface Configuration and Secure Remote Access using Telnet Implementation.

Date of performance: 15.07.26

Date of Submission: 22.07.26

Department of Computer Science and Engineering

# SOUTHEAST UNIVERSITY

Submitted By,

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Section: CSE342.2

Batch: 60

☐ Experiment No: 01

☐ Experiment Name: Basic Router Interface Configuration and Secure Remote Access using Telnet Implementation.

☐ Objective: The Primary focus of this experiment is to secure a Router against unauthorized access by configuring a localized, encrypted authentication database and applying strict transport protocols for remote management. These critical security commands are executed out-of-band via a direct console connection from a end-device. Secondary objective is to include basic IPv4 interface configuration and connectivity verification is other end device.

☐ Network Topology:

- 1x Cisco 2911 Router
- 1x Laptop for Console Connection
- 1x PC
- Console cable
- Cross cable

☐ IP Addressing: ~~Table~~

- Router: 192.68.0.1 255.255.255.0
- PC : 192.68.0.2 255.255.255.0

☐ CLI Command on Laptop: For this we need to open Desktop Interface on Laptop. Then open terminal interface.

Router > enable

Router# configure terminal  To enable Privileged mode.

Router(config)# hostname "RAFIN"  Global Config mode.

Router(config)# enable Password 1111

Router(config)# username RAFIN Password 1111

Router(config)# To write memory

Router(config)# line console 0

Router(config)# login local

|→ enable Password checking

Router(config)# exec-timeout 2 →

Router(config)# service Password-encryption

Router(config)# exit.

☐ To Configure Telnet:

Router(config)# line vty 0 4

Router(config)# transport input telnet

Router(config)# Password 1111

Router(config)# login

Router(config)# exit

### ☐ To Assign IP Address:

Router > enable

Router# configure terminal

Router(config)# interface gig 0/0

Router(config)# ip address 192.68.0.1 255.255.255.0

Router(config)# no shutdown

Router(config)# exit.

### ☐ To Check Connectivity: open Command Prompt Interface on Pe.

- Ping 192.68.0.1 → You should receive four successful replies.

### ☐ Remote Access Test (Telnet):

C:\> telnet 192.68.0.1

Trying 192.68.0.1..... open

User Access Verification

Username: RAFIN

Password:

Router > enable

Password:

Router#

Cisco Packet Tracer - C:\Users\RAFIN\Downloads\NETWORKING LAB - 01.pkt

File Edit Options View Tools Extensions Window Help

Logical Physical x 708, y 101

Root 11:52:30

Basic Router Interface Configuration and Secure Remote Access (Telnet) Implementation

Laptop-PT  
Laptop0

Gig0/0

NET ID: 192.68.0.0

Fa0

PC-PT  
PC0  
192.68.0.2

Name: MD. Kamran Hasan Rafin  
ID: 20220000000066

Time 00:27:33

Realtime Simulation

Scenario 0

New Delete

Toggle PDU List Window

| Fire       | Last Status | Source | Destination | Type | Color | Time(sec) | Periodic | Num | Edit   | Delete   |
|------------|-------------|--------|-------------|------|-------|-----------|----------|-----|--------|----------|
| Successful |             | PC0    | Router0     | ICMP |       | 190.781   | N        | 0   | (edit) | (delete) |

4] Conclusion: This experiment successfully demonstrated secure out-of-band device management and local network configuration using Cisco Packet Tracer. By utilizing the direct console connection on laptop for critical security commands - including encrypted secrets and local user databases were deployed. Explicitly defining VTY transport protocols effectively ~~eliminated~~ eliminated connection closure errors during remote administrative access attempts. Verification tests from PC proved that the router successfully enforced strict credential checks before permitting remote access.